Measuring Spatial Inequality in Iran with Nighttime Lights: Validation, Gas Flaring and the Limits of a Satellite Proxy

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Based on my M.Sc. thesis research in Economics, Sharif University of Technology (supervisor: Dr. Saeed Tajrishy)
Preliminary results, November 2026 · Code and outputs: github.com/arminkhalilzadeh/iran-ntl-regional-inequality

Abstract

Satellite nighttime lights are widely used to measure economic activity where official regional statistics are incomplete. This paper asks how well they work for Iran, a centralised, oil-dependent economy in which official provincial accounts are published with long delays. I build a panel of Iran's 31 provinces for 2012–2024 that combines VIIRS nighttime lights with provincial GDP, census population and household expenditure, and use it to validate lights as a proxy and to measure spatial inequality. Lights track national output reasonably well (elasticity 0.24) and order provinces in a way that agrees moderately with GDP (rank correlations 0.52–0.67). But three problems matter for inequality measurement. First, gas flaring in four oil-producing provinces dominates the level of measured inequality: excluding them lowers the light-based Gini from about 0.39 to 0.25. Second, the remaining series contains a jump in 2017 that is not visible in GDP or expenditure and is more likely a measurement artefact than an economic event. Third, within many provinces, lights rise while real household expenditure falls. Lights are therefore useful for ranking places and for filling gaps in output data, but they should not be read as a measure of welfare. I discuss what these results imply for building a longer series back to 1992.

1. Introduction

Economic policy can target people or places. Cash transfers and subsidies are judged by their effect on the distribution of household welfare, usually measured by a national Gini coefficient. Regional development budgets, infrastructure and industrial zones are judged by their effect on the gap in economic activity across regions. The two dimensions of inequality are related, but they can move in opposite directions: a stable national Gini can hide a widening gap between provinces, and the reverse is also possible. To evaluate place-based policies we need a measure of spatial inequality at the level these policies target (Kanbur and Venables 2005; Novotný 2007). Since Williamson (1965), regional inequality has also been expected to change systematically with development, and Lessmann (2014) finds an inverted-U relationship across countries.

For Iran, such a measure is hard to build from official statistics alone. Provincial GDP is published by the Statistical Centre of Iran with a delay of several years, and the available series is short. At the same time, the shocks that matter most for regional development in Iran, such as sanctions, exchange-rate crises and changes in oil revenue, affect provinces very unevenly, depending on their exposure to trade, industry and oil. This makes Iran an interesting case for spatial inequality, and a difficult one to measure.

Since Chen and Nordhaus (2011) and Henderson, Storeygard and Weil (2012), nighttime lights observed from satellites have become a standard tool for measuring economic activity where official data are weak (Donaldson and Storeygard 2016; Gibson, Olivia and Boe-Gibson 2020). They have even been used to settle disagreements between national accounts and household surveys (Pinkovskiy and Sala-i-Martin 2016). Lessmann and Seidel (2017) used lights to estimate regional income and regional inequality for a large set of countries, and Mveyange (2015) did the same for Africa. For Iran, Farzanegan and Fischer (2021) used lights to study the shadow economy after the lifting of sanctions. The appeal is clear: lights are observed every year, at high spatial resolution, by the same instrument everywhere. The limits are also known. Lights capture some activities better than others, they saturate in dense cities, and they respond to things that are not output, such as gas flares and street lighting (Gibson et al. 2021).

This paper is the measurement part of my M.Sc. thesis. Before using lights to study the evolution of spatial inequality in Iran, I ask whether they measure what they are supposed to measure in this setting. I make three contributions. First, I validate lights against official output at the national and provincial level and against household expenditure from survey data. Second, I show that gas flaring in four oil-producing provinces dominates the level of measured inequality, so that any light-based inequality index for Iran must deal with it explicitly. Third, I document two problems that are less often discussed: a discontinuity in the light series that does not correspond to anything in the economic data, and a negative relationship between lights and real expenditure within many provinces. Together, these results define what lights can and cannot be used for in the rest of the thesis.

2. Data

2.1 Panel structure

The panel covers Iran's 31 provinces within their current boundaries. Keeping boundaries fixed avoids artificial breaks when provinces were split in the past. Iranian statistics follow the solar calendar, in which the year begins in late March; I refer to each Iranian year by the Gregorian year in which it starts, so that 1391 is 2012 and 1403 is 2024.

2.2 Sources

Variable	Definition	Years	Source
Nighttime lights	Sum of annual VIIRS radiance within each province	2012–2024	Google Earth Engine
Provincial GDP	Constant 2021 prices	2012–2021	Statistical Centre of Iran
Population	Census values, interpolated and extrapolated between censuses	2011–2025	Statistical Centre of Iran
Household expenditure	Real expenditure per capita, provincial means	2012–2024	Household Income and Expenditure Survey (HIES)

I extracted annual provincial radiance from VIIRS composites (Elvidge et al. 2017, 2021) in Google Earth Engine, using the provincial shapefile, and divided it by population to obtain lights per capita. Monthly composites were also aggregated to quarters to compare lights with quarterly national GDP. Household expenditure comes from the HIES microdata, deflated by the consumer price index and averaged by province. After extraction, all processing steps are scripted in Python and the regressions are run in Stata.

2.3 Gas flaring

Iran flares large amounts of associated gas at oil and gas fields, and gas flares are among the brightest sources of light in the country; they are visible in VIIRS data and can be located with dedicated detection methods (Elvidge et al. 2016). Four provinces account for most flaring: Bushehr, Khuzestan, Ilam, and Kohgiluyeh and Boyer-Ahmad. These are also among the provinces with the highest GDP per capita, because their output includes oil and gas production. In what follows, I report results for all 31 provinces and for the 27 provinces that remain when these four are excluded. Excluding whole provinces is a blunt solution, and masking flare pixels directly would be better; I return to this point in Section 6.

3. Do Lights Measure Economic Activity in Iran?

3.1 National level

At the national level, I regress log quarterly GDP on the log of total quarterly radiance over 51 quarters. The estimated elasticity is 0.239 (standard error 0.031), with an R-squared of 0.544. For non-oil GDP, the elasticity is almost the same, 0.246 (standard error 0.036). This matters because one might expect lights in Iran to reflect mainly oil and gas activity. The similar elasticity for non-oil output suggests that

lights also capture general economic activity. The size of the elasticity, well below one, is in line with what the literature finds for national aggregates (Henderson, Storeygard and Weil 2012).

3.2 Provincial level

At the provincial level, the question is whether lights order provinces correctly and whether they track changes within provinces over time. These are different properties, and they give different answers.

In a pooled regression of log GDP per capita on log lights per capita across 310 province-years, the elasticity is 0.637 (standard error 0.036, R-squared 0.499). Figure 1 shows the relationship; the correlation is 0.71. Most of this association comes from differences between provinces. When I add province and year fixed effects, so that only changes within provinces are used, the elasticity falls to 0.361 (standard error 0.138, clustered by province, p = 0.014). When the four flare provinces are excluded, it falls further, to 0.227, and is no longer statistically significant (p = 0.127). Lights thus describe which provinces are richer much better than they describe how a given province's output changes from year to year.

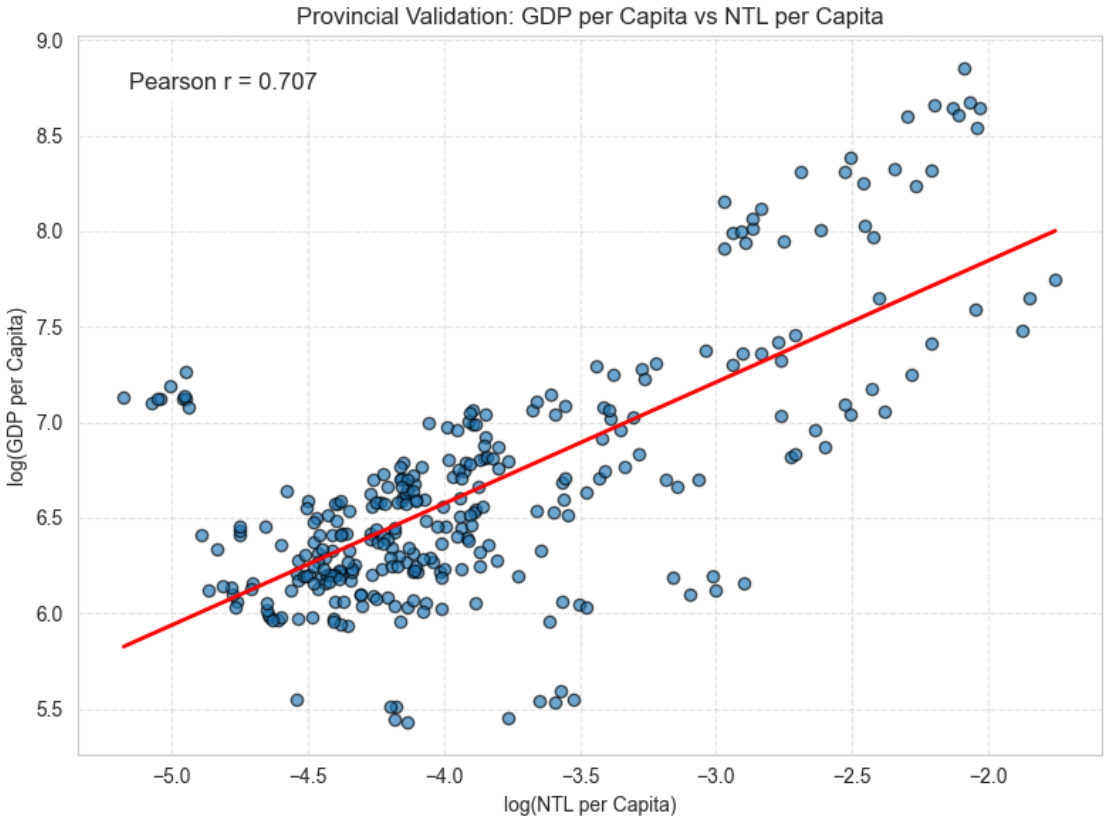


Figure 1. Log GDP per capita against log nighttime lights per capita, 31 provinces, 2012–2021 (310 province-years). Pooled OLS fit; Pearson correlation 0.71.

The growth regressions point in the same direction. Annual growth in lights per capita predicts growth in GDP per capita with a slope of 0.356 (standard error 0.144), but explains only 17.5 percent of the variation across provinces.

3.3 Ranks

Since inequality indices depend on how provinces compare with each other, I also compare ranks. The Spearman correlation between provincial ranks by GDP per capita and by lights per capita lies between 0.52 and 0.67 in every year from 2012 to 2021, and is always significant at the 1 percent level. The deviations, however, follow a clear pattern. In 2021, Tehran ranks 6th in GDP per capita but 31st in lights per capita; Gilan and Alborz also appear much dimmer than their output would suggest. Sistan and Baluchestan, the poorest province by GDP per capita, ranks 10th in lights per capita, and South

Khorasan moves from 27th to 7th. Dense urban provinces produce much output per unit of light, while large, sparsely populated provinces with roads, borders and scattered settlements produce much light per person. This pattern is consistent with what Gibson et al. (2021) report for other countries, and it means that lights per capita should be used with province fixed effects whenever the aim is to predict output.

4. Spatial Inequality, 2012–2024

4.1 Indices

I compute population-weighted inequality indices across provinces for three variables: lights per capita, GDP per capita and real household expenditure per capita. I report the Gini coefficient and, as robustness checks, the coefficient of variation and the generalized entropy family, which includes the mean log deviation and the Theil index (Shorrocks 1980; Cowell 2011; Lessmann and Seidel 2017). Weighting by population means that each index describes inequality between the provinces in which people live, rather than between provinces as units. The expenditure index measures inequality between provincial averages, not between households, and should therefore not be compared directly with the national income Gini.

4.2 The role of gas flaring

Table 1 reports the Gini coefficient for selected years. The first result is that gas flaring dominates the level of measured inequality. With all 31 provinces, the light-based Gini is about 0.39 in 2012 and 0.44 in 2024. Excluding the four flare provinces lowers it to 0.25 and 0.29. GDP-based inequality falls by a similar amount, from 0.36 to 0.23 in 2012, because the same provinces also have the highest measured output from oil and gas. Expenditure inequality, by contrast, hardly changes when these provinces are excluded, because their households are not much richer than households elsewhere.

Year	Lights, all	Lights, excl. flare	GDP, all	GDP, excl. flare	Expenditure, all	Expenditure, excl. flare
2012	0.393	0.245	0.363	0.228	0.159	0.160
2015	0.397	0.242	0.360	0.229	0.188	0.194
2016	0.449	0.219	0.404	0.238	0.195	0.201
2017	0.449	0.289	0.388	0.237	0.189	0.196
2021	0.422	0.298	0.350	0.231	0.200	0.203
2024	0.435	0.294	–	–	0.230	0.236

Table 1. Population-weighted Gini coefficients across provinces. “Excl. flare” drops Bushehr, Khuzestan, Ilam, and Kohgiluyeh and Boyer-Ahmad. Provincial GDP is available until 2021.

The gap between the light-based and the expenditure-based indices is large in both samples. Part of this gap is expected: output is more concentrated than consumption, because of transfers, migration and the location of capital-intensive production. Part of it reflects what lights measure, which I discuss in Section 5.

4.3 Trends

Excluding the flare provinces, the three measures move differently over time (Figure 2). Expenditure inequality between provinces rises steadily, from 0.160 in 2012 to 0.236 in 2024, with a peak of 0.255 in 2023. GDP-based inequality shows no clear trend and stays between 0.22 and 0.24 from 2012 to 2021. Light-based inequality is flat at about 0.24 until 2015, falls to 0.22 in 2016, jumps to 0.29 in 2017 and stays near 0.29–0.30 afterwards.

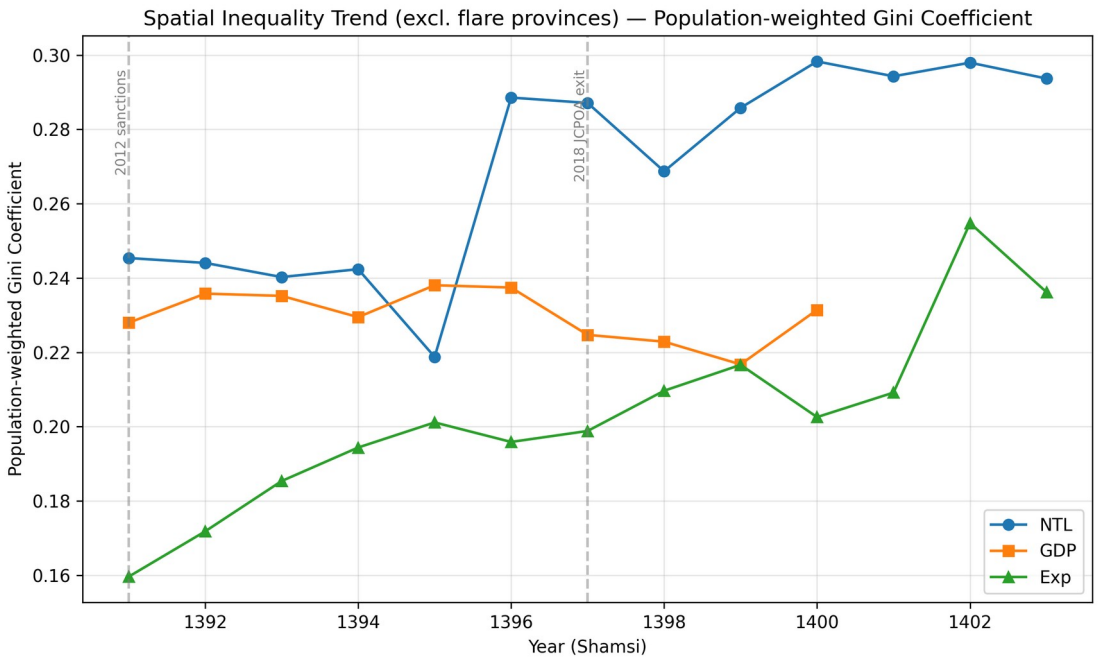


Figure 2. Population-weighted Gini across 27 provinces, excluding the four flare provinces, by Iranian year (1391 = 2012). Lights per capita, GDP per capita and real household expenditure per capita.

If one takes the light series at face value, spatial inequality in Iran increased after 2016, and the increase is in the same direction as the increase in expenditure inequality. Two independent sources moving in the same direction is encouraging. But the timing of the increase in lights is suspicious, and I do not think it should be taken at face value yet.

4.4 Is there a break after 2018?

A natural hypothesis is that the U.S. withdrawal from the nuclear agreement in 2018, and the return of sanctions, changed regional inequality, because sanctions affect trade- and export-oriented provinces more than others. I test for a change in level and in trend after 2018 in each inequality series, using a regression of the index on a linear trend, a post-2018 dummy and its interaction with the trend. For the light-based Gini across all provinces, the trend is positive (0.015 per year, $p = 0.032$), but neither the level shift (0.017, $p = 0.73$) nor the change in slope (−0.011, $p = 0.18$) is significant. The other indices give the same result, and none of the break coefficients is significant when the flare provinces are excluded.

This is a null result, and it should be read carefully. With 13 annual observations and four parameters, the test has little power. I fix the break date at 2018 because it is given by the event; tests with an unknown break date (Andrews 1993) would require a longer series. The correct conclusion is not that sanctions had no effect on spatial inequality, but that annual national-level series are too short to detect such an effect. A better design would exploit differences in exposure across provinces rather than a single break in a national series; I return to this in Section 6.

5. What Lights Do Not Measure

5.1 The 2017 discontinuity

The jump in light-based inequality between 2016 and 2017 (from 0.219 to 0.289 in the sample without flare provinces) has no counterpart in GDP or expenditure, which barely move in that year. An increase of this size within one year would require a large and sudden change in the relative position of provinces, and nothing in the economic data suggests such a change. Two measurement explanations seem more likely. The first is a change in the satellite product: annual VIIRS composites are produced with different versions and processing methods over time (Elvidge et al. 2021), and a change in how background noise

or ephemeral lights are filtered could shift the relative brightness of provinces. The second is the mapping between the Iranian and Gregorian calendars: if monthly composites are assigned to the wrong Iranian year around the change of product, provincial totals would be distorted in exactly one year. The next step is to check both explanations by recomputing the series from a single product version and by testing the calendar mapping month by month. Until this is resolved, I do not interpret the post-2016 increase in light-based inequality as an economic result.

5.2 Lights and welfare within provinces

The second problem concerns the relationship between lights and household welfare. For each province, I estimate separate linear trends in lights and in real expenditure per capita, and compute the within-province correlation between the two series over time. Lights rise in 30 of the 31 provinces. Real expenditure, however, falls in many of them, and in several provinces the within-province correlation between lights and expenditure is strongly negative: −0.78 in East Azerbaijan, −0.69 in Kohgiluyeh and Boyer-Ahmad, −0.63 in Lorestan and −0.60 in North Khorasan. In other provinces the correlation is strongly positive, for example 0.91 in Mazandaran and 0.88 in Tehran.

The most plausible interpretation is that lights in Iran track the expansion of infrastructure, electrification and public lighting, which continued during periods in which real household consumption fell. Lights then measure something like the physical footprint of economic activity rather than living standards. This does not make them useless, but it limits what they can be used for. They are informative about the location and scale of activity, and they can help fill gaps in provincial output where official data are missing. They should not be interpreted as a measure of household welfare, and an increase in lights in a province should not be read as evidence that its residents became better off.

6. Implications and Next Steps

The results so far suggest four lessons for the remainder of the thesis, and for measurement with satellite data more generally.

First, gas flaring must be handled at the level of pixels, not provinces. Dropping four provinces removes not only flares but also a large part of Iran’s oil economy, which changes the question being asked. The next version of the panel will mask flare locations using published flare catalogues (Elvidge et al. 2016) and recompute provincial totals, so that oil-producing provinces can stay in the sample.

Second, the 2017 discontinuity must be resolved before any trend in light-based inequality is interpreted. If it turns out to be a product change, the series will be rebuilt from consistent inputs; if it is a calendar error, it can be corrected directly.

Third, lights are best used for what they do well. The plan is to follow Lessmann and Seidel (2017) and use lights, together with province fixed effects, to predict provincial GDP for the years in which official data are missing, and to combine official and predicted output with optimal weights in the spirit of Henderson, Storeygard and Weil (2012). Backcasting tests, in which the model is estimated on later years and used to predict earlier ones, will show how well this works. Extending the series back to 1992 requires harmonizing the older DMSP satellite data with VIIRS, which is known to be difficult (Li et al. 2020); if harmonization does not produce a consistent series, the main analysis will be limited to the VIIRS period.

Fourth, the question of how shocks such as sanctions affected spatial inequality cannot be answered with a short national series. A more promising approach is to measure each province’s exposure, for example through its share of trade-dependent and export-oriented industry, and to compare the evolution of output across more and less exposed provinces. Here, lights are valuable precisely because they are observed every year for every province, including the years for which official provincial accounts are not yet published.

More generally, the Iranian case shows that a satellite proxy is not a substitute for careful measurement. Lights solve the problem of missing data, but they introduce their own errors, some of which, like gas flaring, are specific to particular economies. Each new source needs to be validated against the others, and its errors need to be understood before it can be used to answer substantive questions.

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